

Ghilmo Choi

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Education

Korea Aerospace University , Republic of Korea B.S. in Mechanical Engineering	Mar 2021 – Feb 2027 (expected)
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- Official GPA: **4.00 / 4.5** (Major GPA: **4.08 / 4.5**)
- Expected GPA after current-semester grades are posted: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
- Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics

Research Interests

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- On-device vision-language models for natural-language robotic mission execution
 - Perception, Prediction, and Decision Making for Autonomous Driving
 - Motion planning and path tracking for real-vehicle autonomous driving

Research / Project Experience

ASOL (Autonomous Systems and Optimization Lab) - <i>Undergraduate Researcher</i>	Korea Aerospace Univ. June 2025 – Present
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- **On-device VLM-based robotic mission execution**
Developed a natural-language mission execution framework that converts user commands into executable robot behaviors on a Traxxas TRX-4 & UGV
- **Hallucination-aware and SLAM-grounded task planning**
Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution

KAUVOY Autonomous Driving Club - <i>Localization Part</i>	Korea Aerospace Univ. Nov 2024 – Nov 2025
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- **EKF-based localization**
Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
- **Autonomous parking and path tracking**
Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid A* path planning, and Pure Pursuit-based path tracking

- <i>Team Leader</i>	Nov 2025 – present
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- **Finite-State-Machine-based decision-making module for autonomous driving scenarios**

Course Project, Introduction to Machine Learning - End-to-End Machine Learning Classification and Regression Pipeline	Korea Aerospace Univ. Aug 2025 – Nov 2025
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- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

Honors & Activities

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- Finalist, **2025 College Mobility Innovation Competition**
 - Excellence Award, **Drone X AI Marketing Challenge** (2026)
 - 2026 - **Airbus 101 Competition** (5th Place)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS `robot_localization` Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git